

static\js\core\charts.js

```
const ChartConfig = {
// Centralized color palette matching application theme and security standards
colors: {
critical: '#f55855',    // Red for critical severity (universal danger color)
high: '#f8a541',       // Orange for high severity (warning/urgency)
medium: '#3b8ded',     // Blue for medium severity (informational)
low: '#42d392',        // Green for low severity (safe/minimal risk)
unknown: '#6b7280',    // Gray for unknown/undefined severity
primary: '#63a4ff',    // Primary accent color for general chart elements
secondary: '#94a3b8'   // Secondary color for supporting elements
},

// Default chart options for consistent appearance across all charts
defaultOptions: {
responsive: true,           // Charts adapt to container size changes
maintainAspectRatio: false, // Allow flexible aspect ratios for different
containers
plugins: {
legend: {
labels: {
color: '#e2e8f0',    // Light color for legend text on dark background
font: { size: 12 }   // Consistent legend font size
}
}
},
scales: {
x: {
ticks: { color: '#94a3b8' },           // Muted color for x-axis labels
grid: { color: 'rgba(255,255,255,0.1)' } // Subtle grid lines
},
y: {
ticks: { color: '#94a3b8' },           // Muted color for y-axis labels
grid: { color: 'rgba(255,255,255,0.1)' } // Subtle grid lines
}
}
},

// Severity colors as array for easy iteration in chart data
severityColors: ['#f55855', '#f8a541', '#3b8ded', '#42d392', '#6b7280'],

// Utility function to create linear gradients for enhanced chart styling
createGradient: function(ctx, color1, color2) {
const gradient = ctx.createLinearGradient(0, 0, 0, 400); // Vertical gradient
gradient.addColorStop(0, color1); // Start color at top
gradient.addColorStop(1, color2); // End color at bottom
return gradient;
}
};

// Chart creation utilities providing factory methods for different chart types
const ChartUtils = {
// Create doughnut-style pie chart for severity distribution visualization
createPieChart: function(ctx, data, options = {}) {
return new Chart(ctx, {
type: 'doughnut', // Doughnut style provides space for central content
data: data,
options: {
...ChartConfig.defaultOptions, // Apply base configuration
cutout: options.cutout || '65%', // Default 65% cutout for doughnut hole
...options // Merge any additional custom options
}
});
},

// Create line chart for trend visualization and time series data
createLineChart: function(ctx, data, options = {}) {
return new Chart(ctx, {
type: 'line',
data: data,
options: {
...ChartConfig.defaultOptions, // Apply base configuration
elements: {

```

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line: { tension: 0.3 },      // Smooth curve without over-smoothing
point: {
  radius: 4,                // Visible points for data clarity
  hoverRadius: 6            // Larger hover area for interaction
},
},
...options                  // Merge any additional custom options
});
},

// Create bar chart for categorical data comparison
createBarChart: function(ctx, data, options = {}) {
  return new Chart(ctx, {
    type: 'bar',
    data: data,
    options: {
      ...ChartConfig.defaultOptions, // Apply base configuration
      ...options                     // Merge any additional custom options
    }
  });
},

// Create radar chart for multi-dimensional security analysis (e.g., CWE analysis)
createRadarChart: function(ctx, data, options = {}) {
  return new Chart(ctx, {
    type: 'radar',
    data: data,
    options: {
      ...ChartConfig.defaultOptions, // Apply base configuration
      scales: {
        r: { // Radial scale configuration
          angleLines: {
            color: 'rgba(255,255,255,0.1)' // Subtle angle lines
          },
          grid: {
            color: 'rgba(255,255,255,0.1)' // Subtle radial grid
          },
          pointLabels: {
            color: '#a9adc1', // Muted color for category labels
            font: {
              size: 13, // Readable font size for labels
              weight: 'bold' // Bold weight for emphasis
            }
          }
        },
        beginAtZero: true, // Start scale from zero for accurate comparison
        ticks: { display: false } // Hide tick marks to reduce clutter
      },
      ...options // Merge any additional custom options
    }
  });
},

```